

AI CONCEPTS REPORT

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1.what is Artificial Intelligence?

Artificial Intelligence is a part of Computer Science that lets machines and computers do things that people usually do. Things like learning and figuring out problems and understanding what people say. Artificial Intelligence also helps machines recognize pictures and make choices. Artificial Intelligence systems use information and special formulas and machine learning to get better at what they do, over time. Artificial Intelligence does this to improve how Artificial Intelligence works.

How Artificial Intelligence Works:

Artificial Intelligence systems use an important things:

- **Data.** This is the information that Artificial Intelligence uses to learn and get better
- **Algorithms.** These are like rules that tell the system how to look at the data and what to do with it
- **Machine Learning.** This is a way that computers can learn from the data and get better at what they do over time

There are a few types of Artificial Intelligence:

Artificial Intelligence can be broken down into a categories.

- There is Narrow Artificial Intelligence. This is designed to do one thing like filter out spam emails or work as a virtual assistant.
- Then there is General Artificial Intelligence. This is an idea for an Artificial Intelligence that could do anything a human can do.
- And finally there is Super Artificial Intelligence. This is an Artificial Intelligence that would be even smarter, than humans.



2. Difference between AI, Machine learning, Generative AI, and Agentic AI?

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)	Generative AI (GenAI)	Agentic AI
Definition	Broad field of creating systems that mimic human intelligence	Subset of AI that learns from data	Subset of ML that creates new content	AI that can plan, decide, and act toward goals
Main Purpose	Solve intelligent tasks	Learn patterns and make predictions	Generate new content	Complete tasks autonomously
Learns from Data?	Sometimes	✓ Yes	✓ Yes	✓ Usually
Generates Content?	Sometimes	Rarely	✓ Yes	✓ Yes
Makes Decisions?	Basic rules or logic	Based on learned patterns	Limited	✓ Advanced decision-making
Plans Multi-Step Tasks?	✗ No	✗ No	Limited	✓ Yes
Uses External Tools?	Usually not	Usually not	Sometimes	✓ Frequently
Acts Autonomously?	✗ No	✗ No	✗ No	✓ Yes
Human Involvement	High	Moderate	Moderate	Low (with supervision)
Typical Output	Decisions, classifications, actions	Predictions, recommendations	Text, images, code, audio, video	Actions, workflows, task completion
Examples	Chess AI, route planning, voice assistants	Spam filters, recommendation engines, fraud detection	ChatGPT, DALL·E, Mid journey, GitHub Copilot	Autonomous assistants, AI agents that book meetings, manage projects, or perform research
Real-World Analogy	A smart machine	A student learning from examples	A creative artist	A project manager who plans and executes tasks

3.Explain any three open-source AI models available on Hugging Face?

1. Llama 3

Llama 3 is a language model that people can use for free. It was made by Meta AI. This Llama 3 model is really good at understanding what people say and writing things that sound like a person wrote them. It can do lots of things like answer questions write text help with coding and have conversations. People who study and make things with computers like Llama 3 because it is very good and can be taught to do tasks.

2. Mistral 7B

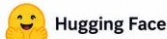
Mistral 7B is another language model that people can use for free. It was made by Mistral AI. Even though it is not as big as some models it still works very well and does not need as much computer power. People use Mistral 7B to generate text summarize things, translate languages and make chatbots. It is a choice for people who do not have very powerful computers.

3. Stable Diffusion

Stable Diffusion is a kind of model that makes pictures from what people write. It was made by Stability AI. If you tell it something like "a city in the future at sunset" it will make a picture of that. People use Stable Diffusion to make art, design graphics make ads and create content. It is very good at making pictures quickly and can be set up to make pictures in different styles.

3 OPEN-SOURCE AI MODELS ON HUGGING FACE

Powerful models. Open access. Limitless possibilities.





Llama 3

by Meta AI



Llama 3 is a powerful open-source large language model designed for understanding and generating human-like text.

Key Features

- Strong reasoning and conversation abilities
- Excellent for coding and complex tasks
- Available in multiple sizes (8B, 70B)

Common Applications

- Chatbots
- Content Generation
- Coding Assistance
- Question Answering



Mistral 7B

by Mistral AI



Mistral 7B is a lightweight yet highly efficient language model with 7 billion parameters.

Key Features

- High performance with lower computational cost
- Efficient and fast inference
- Ideal for real-world applications

Common Applications

- Text Generation
- Summarization
- Translation
- Virtual Assistants



Stable Diffusion

by Stability AI



Stable Diffusion is an open-source text-to-image model that generates high-quality images from text descriptions.

Key Features

- High-quality image generation
- Customizable and fine-tunable
- Supports various styles and resolutions

Common Applications

- Digital Art Creation
- Graphic Design
- Image Editing
- Content Creation

Why Open-Source Models?



Free Access
Most models are free to use and modify.



Customization
Fine-tune models for specific tasks.



Community Support
Large ecosystem and active community.



Transparency
Open weights and documentation promote trust and research.

💡 These models are available on Hugging Face 🤗 and can be used for research, education, and commercial applications.

4.what is the difference between traditional programming and AI-assisted development?

Aspect	Traditional Programming	AI-Assisted Development
Definition	Developers write all code manually using programming languages and predefined logic.	Developers use AI tools to help generate, suggest, optimize, and debug code.
Development Process	Human writes, tests, and maintains code.	AI assists with coding while humans guide and review the output.
Code Creation	Entirely manual.	AI can generate code from natural language prompts.
Problem Solving	Programmer defines every rule and instruction.	AI can suggest solutions based on patterns learned from large codebases.
Speed	Generally slower for repetitive tasks.	Faster due to automated code generation and suggestions.
Debugging	Developers identify and fix errors manually.	AI can detect bugs and recommend fixes.
Learning Curve	Requires deep knowledge of syntax and frameworks.	AI helps beginners by providing examples and explanations.
Productivity	Depends entirely on developer expertise and effort.	Higher productivity through automation of routine tasks.
Accuracy	High when written and reviewed carefully.	Can be high, but AI-generated code must still be verified.
Examples	Writing a sorting algorithm from scratch.	Asking an AI assistant to generate a sorting algorithm.

5.Explain AI hallucinations with one example?

AI Hallucinations

AI hallucinations happen when an AI system makes up information that sounds right but is actually wrong or not true.

This happens because AI models try to guess the answer based on what they have learned from their data. They do not check facts like people do.

Example

User: Who invented the smartphone in 2010?

Hallucinated AI Response:

"John Smith invented the smartphone in 2010."

This answer is wrong because there is no one named John Smith who invented the smartphone. The AI just made up a person and a fact that sound right but are not.

Why Do Hallucinations Occur?

- The input is not clear. Is incomplete.
- The AI does not have the information.
- The AI predicts what is likely to be said of checking facts.
- The AI does not have data, on a topic to give a correct answer.

